Card fraud triage

Learning normal activity with a GMM

Our goal: decide which transactions a person should review first.

Phase 1 only. Team T011, Section A, Project 26.

Literature behind the method

EM explains fitting the hidden component memberships.

PR measures expose the burden of false alerts when fraud is rare.

AIC and BIC balance legitimate density fit against extra parameters.

Our results come from the experiment, not from another paper.

The data and the three time periods

283,726 transactions after removing exact repeats. Class 0 = legitimate, 1 = fraud.

Period	Transactions	Frauds
Training	170,235	342
Validation	56,745	57
Test	56,746	74

Only 169,893 legitimate training rows teach the model.

The 29 inputs

V1-V28 plus LogAmount = log(1 + Amount)

Example of the transform: Amount 9 becomes 2.30; 999 becomes 6.91.

Scale each input using the legitimate training mean and standard deviation.

Keep Class out of the inputs. Use Time for splitting.

How the GMM learns

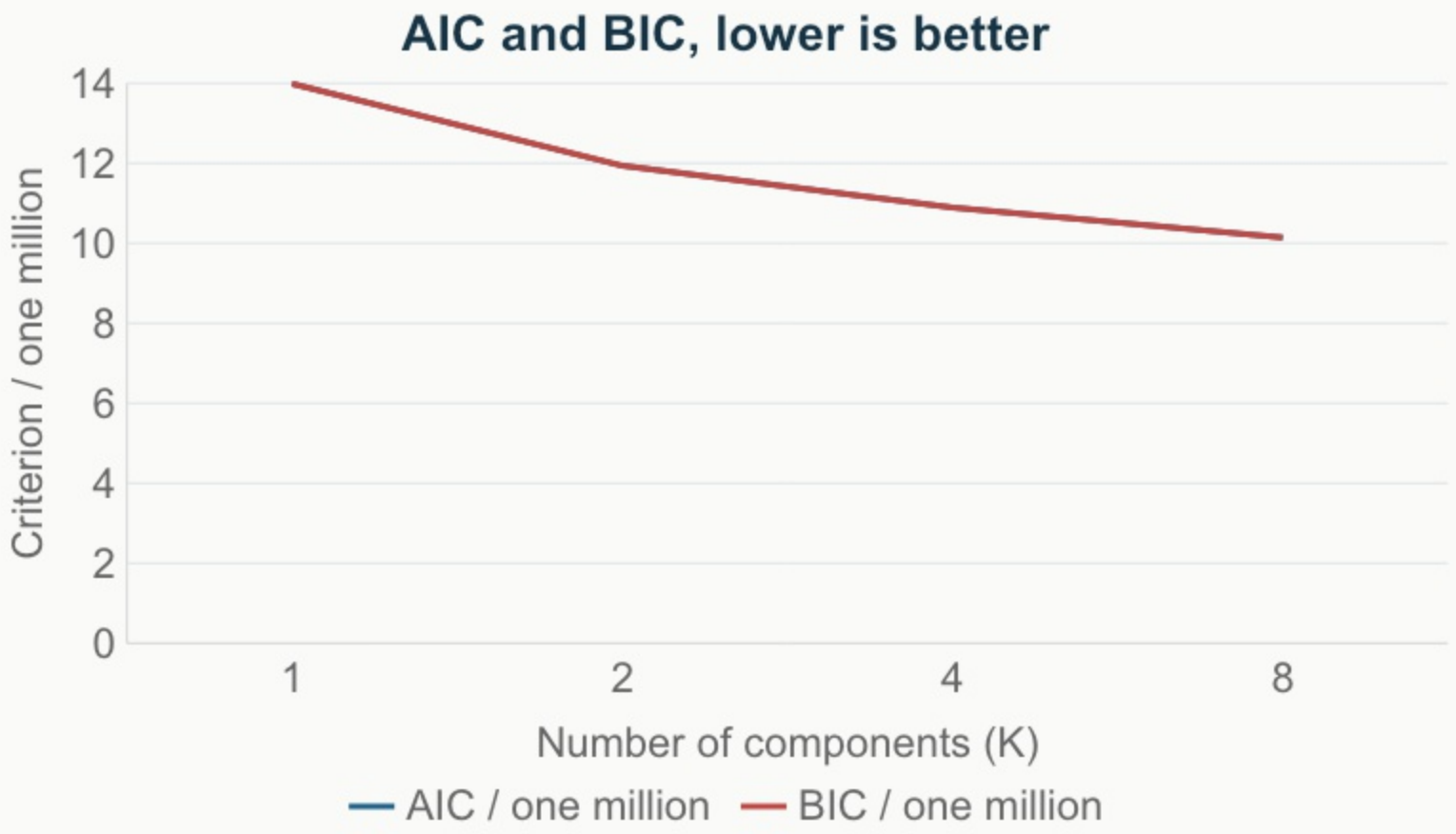
A Gaussian describes a bell-shaped pattern with a center and spread.

A mixture combines several patterns. K is the number of components.

E-step: estimate membership in each component.

M-step: update centers, spreads and weights. Repeat.

Choosing the number of components



Try 1, 2, 4 and 8.

Choose the smallest BIC.

Eight wins in this grid.

From a score to a review alert

Anomaly score = -log density

A larger score means the transaction is less typical.

Default cutoff: 75.8942, chosen on validation data.

Rule examples: score 80 gets an alert; score 60 does not.

The default test result

Actual / decision	No alert	Alert
Legitimate	56,500	172
Fraud	37	37

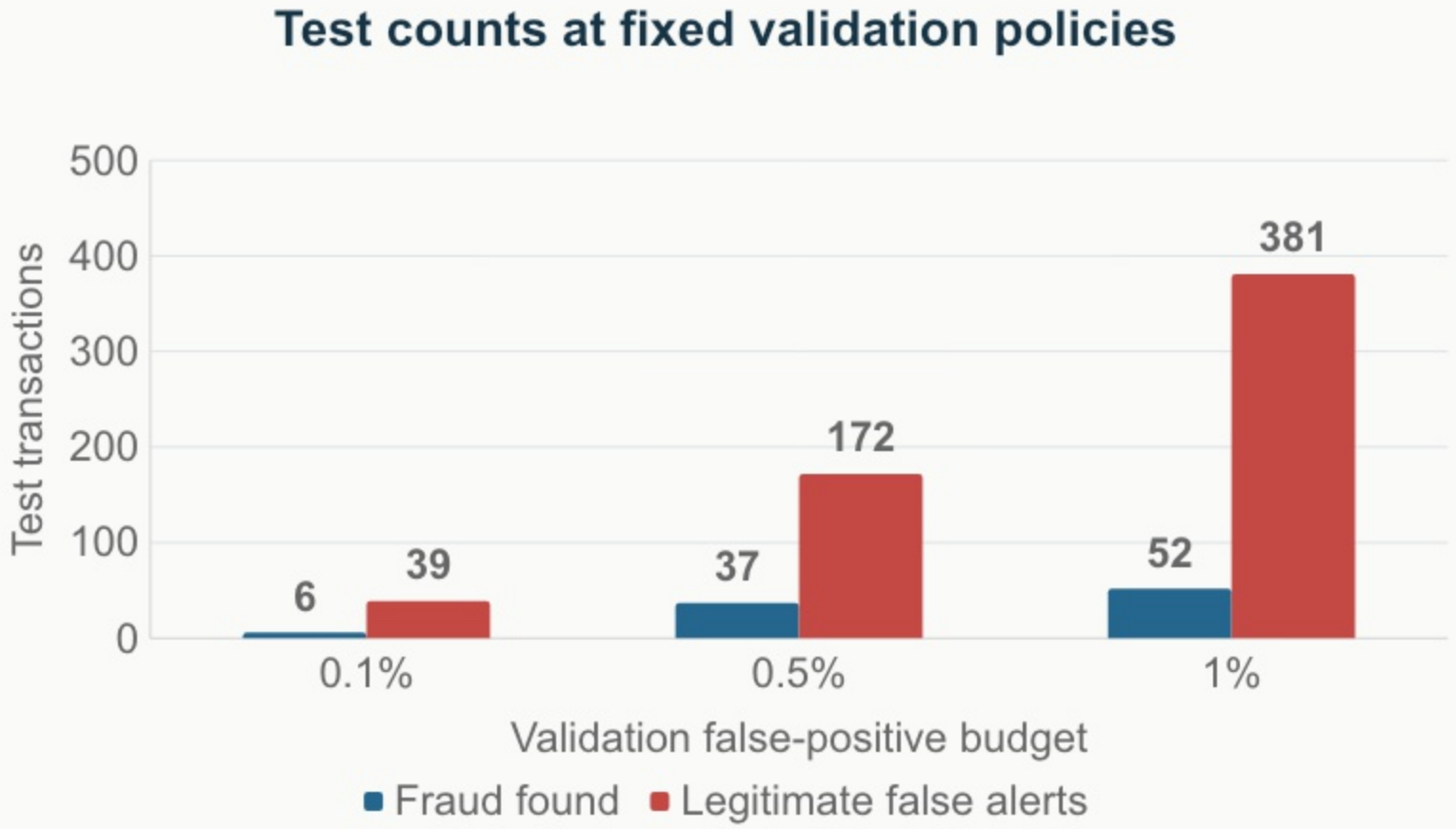
Recall: 37 / 74 = 50%

Precision: 37 / 209 = 17.70%

FPR: 172 / 56,672 = 0.3035%

209 reviews. We find 37 frauds and miss 37.

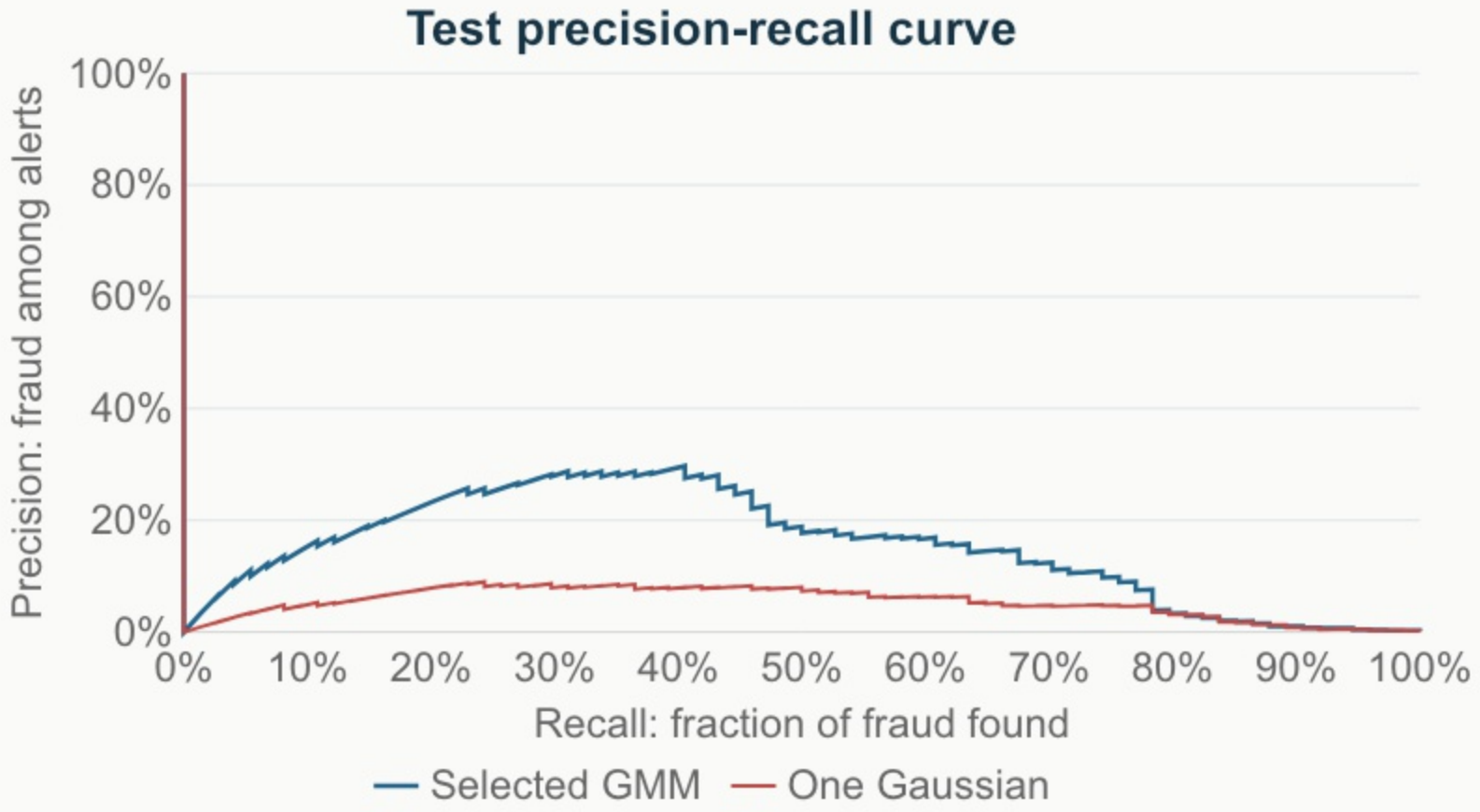
More review brings more false alerts



A looser cutoff catches more fraud here.

It also sends more legitimate transactions for review.

Ranking quality and 100 reviews



PR AUC
GMM: 0.1496
One Gaussian: 0.0527

Top 100 reviews:
29 frauds, 71 legitimate
Precision@100 = 29%

Reproducing and explaining the project

Read 01. Run 02 to 05 in order: restart kernel, then Run All.

Checks cover the time split, scaler fit, convergence and validation budgets.

The checklist links all nine Phase 1 areas to evidence.

Every member must explain the full project and record actual contributions.

What we can conclude

The GMM gives us a clear order for reviewing transactions.

At our default cutoff, it finds half the test fraud and creates 172 false alerts.

Only 74 test frauds. Short historical data. Hidden feature meanings.

Phase 1 covers the GMM. The autoencoder belongs to Phase 2.